

Your Paper Title Here

Your Name(s)

Date

Abstract

Write your abstract here.

Keywords

Keyword 1, Keyword 2, Keyword 3, Keyword 4

Introduction / Background

Provide an introduction with background information.

Statistical Methods

Model

Describe the statistical model used.

Likelihood Inference

Detail the likelihood approach.

Bayesian Inference

Detail the Bayesian approach.

Let's say we believe

$$P(\psi \geq 0.3) = P\left(\frac{1-2\pi}{1-\pi} \geq 0.3\right) = 0.1 \rightarrow P\left(\pi \leq \frac{7}{17}\right) = 0.1$$
$$P(\psi \geq 0) = P\left(\frac{1-2\pi}{1-\pi} \geq 0\right) = 0.1 \rightarrow P\left(\pi \leq \frac{1}{2}\right) = 0.5$$

```
library(LearnBayes)

beta.select(quantile1 = list(p = 0.1, x = 7/17),
            quantile2 = list(p = 0.5, x = 0.5))
```

```
## [1] 26.2 26.2
```

That is,

$$P(\pi|x) \sim \text{Beta}(26.2 + 8, 26.2 + 162)$$

$$\begin{aligned}
P(\Psi \leq \psi|x) &= P\left(\frac{1-2\pi}{1-\pi} \leq \psi|x\right) \quad -\infty < \psi \leq 1 \\
&= P\left(\pi \geq \frac{\psi-1}{\psi-2}|x\right) \\
&= 1 - P\left(\pi \leq \frac{\psi-1}{\psi-2}|x\right) \\
F_{\Psi}(\psi|x) &= 1 - F_{\pi}\left(\frac{\psi-1}{\psi-2}|x\right) \\
f_{\Psi}(\psi|x) &= \left(1 - F_{\pi}\left(\frac{\psi-1}{\psi-2}|x\right)\right)' \left(\frac{\psi-1}{\psi-2}\right)' \\
f_{\Psi}(\psi|x) &= f_{\pi}\left(\frac{\psi-1}{\psi-2}|x\right) \frac{1}{(\psi-2)^2} \\
&= \frac{\left(\frac{\psi-1}{\psi-2}\right)^{33.2} \left(\frac{-1}{\psi-2}\right)^{187.2}}{B(34.2, 188.2)} \frac{1}{(\psi-2)^2} \\
&= \frac{(1-\psi)^{33.2}}{B(34.2, 188.2)(2-\psi)^{222.4}} \quad -\infty < \psi \leq 1
\end{aligned}$$

The prior for Ψ will be

$$f_{\Psi}(\psi) = \frac{(1-\psi)^{25.2}}{B(26.2, 26.2)(2-\psi)^{52.4}} \quad -\infty < \psi \leq 1$$

```

ggplot()+

  geom_function(

    fun = function(x) {(1-x)^33.2/(beta(34.2, 188.2)*(2-x)^222.4)},

    xlim = c(0, 1),

    aes(color = 'Poserior'),

    lwd = 1.2

  )+

  geom_function(

    fun = function(x) {(1-x)^25.2/(beta(26.2, 26.2)*(2-x)^52.4)},

    xlim = c(0, 1),

    aes(color = 'Prior'),

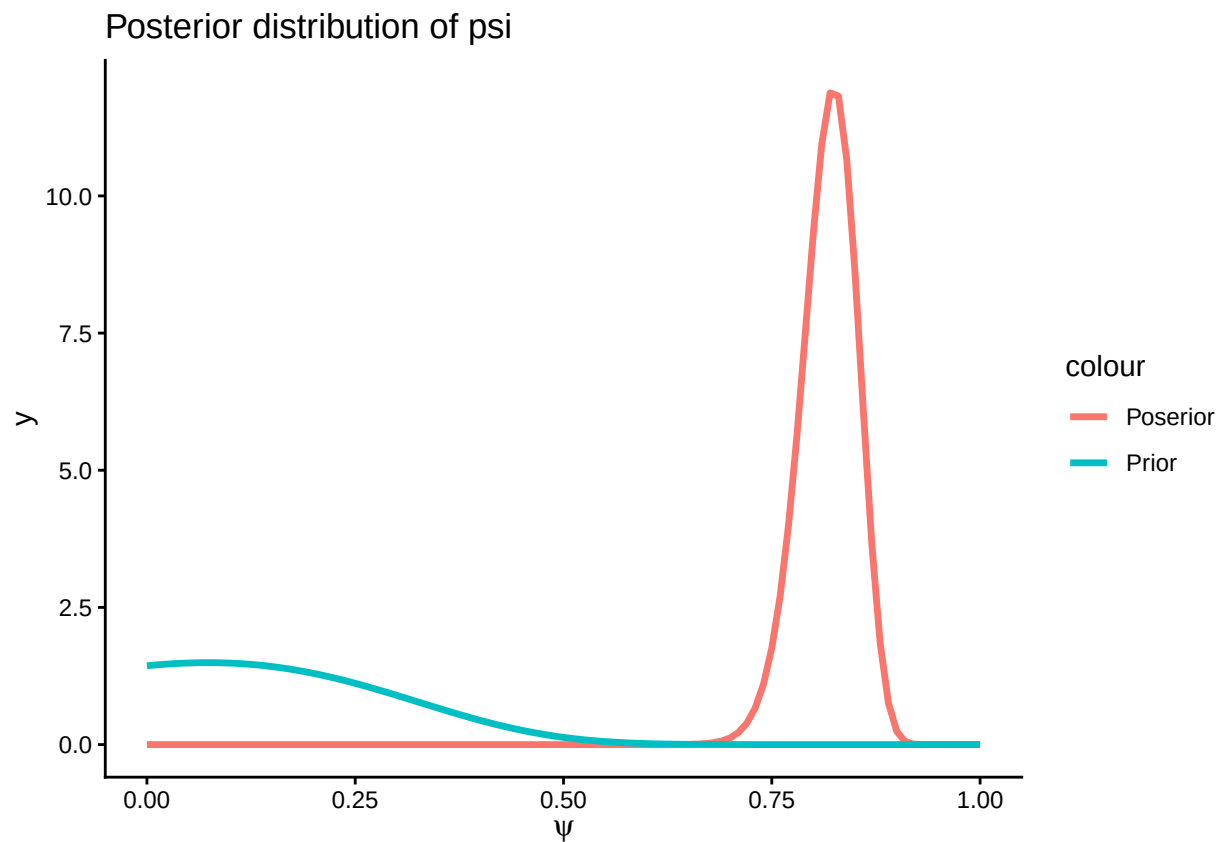
    lwd = 1.2

  )+

  labs(x = expression(psi), title= paste('Posterior distribution of psi') )+

```

```
theme_classic()
```



Results

To perform inference

Present your findings.

Median of posterior distribution:

$$P(\psi \leq q|u)$$

$$P(1 - 2\pi \leq (q|u)(1 - \pi))$$

```
inverse <- qbeta(0.5, 34.2, 188.2, lower.tail = F)
q <- (2*inverse - 1)/(inverse - 1)
```

To perform inference with Bayesian methods, summary statistics of the posterior distribution must be reported. For this report, the posterior median will be reported.

To find the posterior median,

$$\begin{aligned}
P(\psi \leq q|v_{obs}) &= 0.5 \\
P\left(\frac{1-2\pi}{1-\pi} \leq q|v_{obs}\right) &= 0.5 && \text{Definition of } \psi \\
P(1-2\pi \leq q_v - q_v\pi) &= 0.5 \\
P(1-q_v \leq 2\pi - q_v\pi) &= 0.5 \\
P\left(\frac{1-q_v}{2-q_v} \leq \pi\right) &= 0.5 \\
1 - P\left(\frac{1-q_v}{2-q_v} \geq \pi\right) &= 0.5 && \text{Definition of complement} \\
\frac{1-q_v}{2-q_v} &\approx 0.1527383 && \text{Inverting probability function} \\
(1-q_v) &= (2-q_v)(0.1527383) \\
q_v &= \frac{2 * 0.1527383 - 1}{0.1527383 - 1} \\
q_v &= 0.8197271
\end{aligned}$$

The posterior median of Pfizer's COVID-19 efficacy, ψ , is 0.819271. To fully contextualize this, the upper and lower bounds of the credible interval are: Lower Bound = $q_v = \frac{2*0.1095326-1}{0.1095326-1}$, 0.1095326 is 0.005 percentile of posterior distribution.

Upper Bound = $q_v = \frac{2*0.2039136-1}{0.2039136-1}$, 0.2039136 from 99.5 percentile of posterior distribution.

```

lower_q = qbeta(0.005, shape1 = 34.2, shape2 = 188.2)
upper_q = qbeta(0.995, shape1 = 34.2, shape2 = 188.2)

lowbound <- (2*lower_q-1)/(lower_q - 1)
upbound <- (2*upper_q - 1)/(upper_q - 1)

```

Our credible interval is [0.8917394,0.7154829]. This means there is a 99% probability that the real value of ψ lies within this interval.

For the frequentist method,

Discussion / Conclusion

Discuss / conclude here.

Bibliography

Brown, B. (2024). *Lecture Title*. Lecture slides, Course Name, University Name.

Doe, J. (2020). Title of the Paper. *Journal Name*, 12(3), 45-67.

Last, F., & Last, F. (2025). *Book Title*. Publisher.

Smith, A., & Johnson, C. (2023). *Title of the Online Article*. Retrieved from <https://www.example.com>.

Appendix

Code

You can reference your code in the appendix (sample here).

Proofs

If applicable, include detailed mathematical derivations or additional theoretical explanations.